# Wind-Assisted Propulsion on North Sea Routes

Maritime Performance Analytics for Flettner Rotors

University project report

May 2026

Aleksandra Kłos
Olga Grigorieva
Elen Mouradian
Bartosz Maj
Hubert Jaczyński

# Contents

# 1 Executive summary

The project studies how wind, waves, currents, and vessel heading influence operational speed on North Sea routes, and whether Flettner rotors can improve voyage performance or enlarge useful weather windows. The work follows the extended university project scope: AIS position reports were merged with ERA5 wind fields and Copernicus Marine sea-state/current variables [3, 4], transformed into trajectory-aware features, modelled with leakage-aware train/test splits, and used in a rotor-assisted what-if simulation.

The final engineered dataset contains 125,160 AIS-metocean observations from 528 voyages between 2023-02-02 and 2026-02-02. The speed target is SOG in knots. The modelling split is chronological by voyage: 422 voyages and 88,346 rows are used for training; 106 later voyages and 35,758 rows are held out for testing. This design prevents observations from the same voyage from appearing on both sides of the split and avoids temporal leakage.

Four model levels were evaluated. A constant-speed baseline gives MAE 1.388 kn and RMSE 1.893 kn. A linear model using wind, wave height, and encoded wind angle gives a small improvement, MAE 1.377 kn and $R^2 = 0.034$. A weather-only XGBoost model is useful for interpretation but does not improve test accuracy under the current feature set. The best predictive model is the lagged nowcast XGBoost model, which adds previous observed SOG and apparent wind context; it reaches MAE 0.406 kn, RMSE 0.758 kn, $R^2 = 0.839$, and 91.7% of predictions within $\pm 1$ kn.

The rotor scenario is deliberately applied after prediction rather than used as a training feature. Rotor power is estimated from the provided polar diagram and converted to a speed increment with the scenario assumption 200 kW $\approx$ 1 kn near the operating point. On the held-out test set, the rotor is active in 98.1% of observations, with mean active rotor power 41.2 kW, mean active speed contribution 0.206 kn, and overall mean contribution 0.202 kn. Under the weather-window definition SOG $\geq 10$ kn and $H_s \leq 3$ m, coverage increases from 76.0% in the baseline prediction to 77.5% after adding the rotor scenario. The number of separate windows decreases from 866 to 808, while points inside windows increase by 529, suggesting that some marginal windows become longer and merge when rotor assistance is applied.

Because the historical AIS period may already include rotor-assisted operation, the scenario should be read as a transparent rotor-opportunity estimate rather than as a measured causal before/after effect.

The optional extensions add operational interpretation. The energy model estimates 112.2 t of fuel saved and 349.4 t $CO_2$ avoided over the three-year dataset, or about 37.4 t fuel and 116.4 t $CO_2$ per year. Route segmentation shows different benefits by heading regime, with westbound legs gaining the largest weather-window coverage increase. Route optimisation demonstrates a Dijkstra grid-routing proof of concept for rotor gain. Gain detection shows that only 11.9% of observations are high-gain moments, but those moments carry a large share of the total benefit; high gain is associated mainly with wind speeds above 8–12 m/s and beam-to-quartering apparent wind angles. The latest Monte Carlo routing extension adds uncertainty-aware route comparison: for Voyage 364, the expected-cost route lowers mean rotor-assisted fuel from 50.72 t on the observed route to 47.81 t under the sampled metocean scenarios, and beats the shortest feasible route in 85.6% of samples.

**Table 1:** Main quantitative outcomes from the analysis run.

| **Area** | **Current result** |
|---|---|
| Dataset | 125,160 observations, 528 voyages, 2023-02-02 to 2026-02-02. |
| Validation | Chronological voyage split; 422 train voyages and 106 test voyages; no temporal leakage. |
| Best speed model | XGBoost lagged nowcast: MAE 0.406 kn, RMSE 0.758 kn, $R^2 = 0.839$. |
| Rotor speed uplift | Mean active $\Delta$SOG 0.206 kn; overall mean $\Delta$SOG 0.202 kn. |
| Weather windows | Coverage 76.0% for the baseline prediction and 77.5% after adding the rotor scenario; +529 test observations inside windows. |
| Fuel and emissions | Scenario estimate of 112.2 t fuel saved and 349.4 t $CO_2$ avoided over the dataset; annualised 37.4 t fuel and 116.4 t $CO_2$. |
| High-gain moments | 11.9% of observations have $\Delta$SOG $\geq 0.5$ kn; classifier F1 = 0.992. |
| Monte Carlo routing | Expected-cost route for Voyage 364: 47.81 t mean rotor-assisted fuel, 46.97–48.64 t 5–95% interval, and 85.6% probability of beating the shortest feasible route. |

## 2 Project context and objectives

Commercial vessels increasingly use wind-assisted propulsion to reduce fuel burn, emissions, and exposure to volatile fuel prices. A Flettner rotor is a rotating vertical cylinder. When wind flows around it, the Magnus effect creates an aerodynamic lift force. Depending on the apparent wind angle and the vessel's course, part of this force can point forward and support the main engine. In practice the contribution is highly conditional: the same rotor can be useful, neutral, or even unfavourable depending on wind speed, angle, sea state, vessel speed, and operational constraints. This conditionality is consistent with published route-based and ship-performance studies of Flettner rotors and other wind-assisted propulsion technologies [5, 6].

The university project frames the problem as an algorithmic analysis of vessel traces and metocean conditions. The expected outputs are dependencies between environmental conditions and voyage parameters, a scenario comparison between baseline and rotor-assisted operation, and an assessment of weather windows for sailing. The extended analysis includes current/tide vectors, apparent wind calculation, rotor modelling, weather-window estimation, and uncertainty analysis.

The practical goal is not to build a final naval-architecture simulator. Instead, the goal is to create a transparent decision-support prototype that can answer questions such as:

- How accurately can SOG be predicted from forecast-available metocean variables and recent vessel behaviour?
- Which environmental variables appear to influence speed under the current data and modelling setup?
- Under what wind-speed and wind-angle combinations do rotors provide meaningful operational gain?
- Do rotors increase the fraction of time in which a voyage satisfies a useful weather-window criterion?
- Can energy, $CO_2$, route segmentation, routing, and uncertainty analyses be built consistently on top of the same feature set?

This framing is important because wind-assisted propulsion is not a single constant correction to vessel speed. A rotor does not simply add the same number of knots in all conditions. It depends on the direction and strength of the wind seen by the ship, the ship's course, the current, and the operational state of the vessel. A model that ignores those dependencies may still produce a number, but it would not be useful for route planning or for deciding when the rotor is actually valuable.

The wider motivation is also regulatory and environmental. The IMO greenhouse gas studies identify international shipping as a material source of global emissions and provide a common reference point for evidence-based reduction measures [1].

The project therefore treats the problem as a chain of decisions. First, the trajectory has to be reconstructed and enriched with environmental context. Second, speed has to be predicted in a way that respects voyages and time. Third, a rotor scenario has to be applied consistently to the same observations. Finally, the result has to be translated into operational quantities: weather windows, fuel, $CO_2$, route segments, and high-gain moments. Each stage is deliberately kept visible in the notebooks so that a reviewer can see where the numbers come from.

There is also an interpretability goal. A purely black-box result such as "rotors improve performance" would not be enough for a maritime user. The operator needs to know when the improvement appears, what assumptions created it, and whether the result is stable enough to support planning. For that reason the report separates predictive accuracy from physical interpretation. The lagged model answers a practical nowcasting question. The weather-only and rotor-gain analyses answer a different question: what environmental structure is associated with better or worse performance?

### 2.1 Operational meaning of the problem

The operational problem can be described from the perspective of a voyage planner. Before a voyage, the planner has a forecast of wind, waves, and currents, a desired route, and a schedule. During the voyage, the operator also has recent vessel speed and a clearer view of how the ship is behaving. The project supports both moments. The weather-only features are closer to the pre-voyage planning situation, while the lagged nowcast model is closer to an in-voyage monitoring situation.

Weather windows are a natural way to turn model output into an operational language. A ship does not only need a high average speed; it needs continuous periods in which speed and sea state are acceptable. A short improvement may be irrelevant if it happens during an already safe and fast period. A small speed increase may be valuable if it connects two otherwise separated windows, keeps a voyage above a schedule threshold, or allows the main engine to run at a lower load while preserving the same arrival time.

This is why the report tracks both speed-prediction metrics and window metrics. Regression metrics show whether the model can reproduce observed speed. Window metrics show whether the predicted difference changes the operational classification of a route segment. The two are related but not identical. A model can have a modest average speed improvement and still be useful if the improvement occurs at the threshold where decisions are made.

## 2.2 Key concepts

Several terms appear repeatedly in the report. SOG means speed over ground: the speed of the vessel relative to the Earth, including the effect of current. COG means course over ground: the direction in which the vessel is moving on the map. These are AIS-derived movement quantities, not direct measurements of engine load or speed through water.

**Table 2:** Short glossary for the main operational and modelling concepts.

| Concept | Meaning in this project |
|---|---|
| SOG | Observed or predicted vessel speed over ground, measured in knots. |
| COG | Vessel movement direction over ground, used to define route context and relative angles. |
| $H_s$ | Significant wave height, used both as a sea-state feature and as part of the weather-window threshold. |
| True wind | Wind from the gridded metocean product, before correcting for vessel motion. |
| Apparent wind | Wind experienced by the moving vessel; especially important for rotor behaviour. |
| Relative wind angle | Angle between wind direction and vessel course; one of the main drivers of rotor usefulness. |
| Weather window | A continuous period where the predicted speed and wave state satisfy an operational threshold. |
| High-gain moment | An observation where the rotor scenario gives a speed contribution of at least 0.5 kn. |

This terminology also explains why the project uses vector and angle features instead of only scalar wind and wave values. A wind speed of 10 m/s can be helpful, neutral, or unfavourable depending on its direction relative to the ship. A current of the same magnitude can support the ship, oppose it, or push it sideways. The model therefore needs both magnitudes and directions to represent maritime performance in a way that is useful for decision-making.

# 3 Data sources and preprocessing

## 3.1 Raw inputs

The core input is a set of AIS-derived position reports for the analysed vessel. Each point contains time, latitude, longitude, speed over ground, course over ground, and trajectory-derived distance fields. These observations are enriched with gridded metocean data from numerical models:

- ERA5 wind variables: 10 m wind components and gust information.
- Copernicus Marine sea state: significant wave height, mean wave direction, and mean wave period.
- Copernicus Marine currents: eastward and northward current components.

AIS is used here as a standardised vessel movement trace; the technical characteristics of the underlying VHF automatic identification system are defined by ITU-R Recommendation M.1371 [2]. The environmental context follows established reanalysis and operational ocean-data sources: ERA5 for atmospheric fields [3] and Copernicus Marine for marine waves and currents [4].

The merged `MasterSet.csv` file contains the base AIS-metocean table. The engineered feature file contains the modelling features used by the baseline and extension notebooks. The feature-engineering notebook reports 125,160 rows and 57 columns after processing.

AIS data should be interpreted as an operational trace rather than a laboratory measurement. It tells us where the vessel was and how fast it moved over the ground, but it does not fully explain why it moved that way. Speed can be affected by engine settings, safety margins, traffic, port approach rules, cargo state, draught, fouling, and schedule decisions. Some of those factors are not present in the dataset. This is the main reason the project keeps a simple constant-speed baseline and a linear weather baseline in the evaluation: they show how much the more complex models improve over a minimal explanation.

The environmental data provide the surrounding physical context. ERA5 wind fields are used to describe the air flow that drives the rotor scenario and affects resistance. Copernicus wave variables describe sea-state severity, which is important because high waves can reduce speed or make a route operationally unattractive even if the wind is favourable. Current vectors are included because the target is speed over ground. A following current and an opposing current can change observed ground speed even if the vessel's speed through the water is similar.

The final modelling table is therefore a joined view of two different worlds: the ship trajectory and the gridded environment. The value of the table comes from the fact that every row is an interpretable event: at this time, in this position, on this voyage, the vessel had this course and speed, while the surrounding wind, wave, and current conditions had these values.

## 3.2 Spatio-temporal alignment

AIS points are irregular in time and space, while metocean products are gridded. The preprocessing pipeline aligns these sources by interpolating environmental variables to each AIS position and timestamp. This is the key step that turns a vessel trajectory into a supervised learning table. For each AIS observation the pipeline attaches the environmental state that the vessel experienced at that location and time.

The workflow separates this work across proof-of-concept interpolation, ERA5-checking, exploratory-analysis, and feature-engineering notebooks. The resulting dataset spans 2023-02-02 to 2026-02-02 and covers 528 voyages.

The alignment step is especially important for wind-assisted propulsion. A small time mismatch can change the estimated wind angle, and a spatial mismatch can attach weather from the wrong grid cell. For ordinary exploratory analysis, that error may only add noise. For rotor analysis, it can change whether an observation is classified as favourable or unfavourable. This is because rotor power is strongly directional: the difference between near-beam wind and near-headwind can be the difference between positive useful power and almost no benefit.

The project also treats voyages as groups rather than as a single continuous stream. This avoids mixing different trips and prevents lag features from borrowing information across voyage boundaries. It also makes evaluation more realistic. A future voyage is not just another random row from the same trip; it is a new sequence of points with its own route, season, weather, and operating pattern.

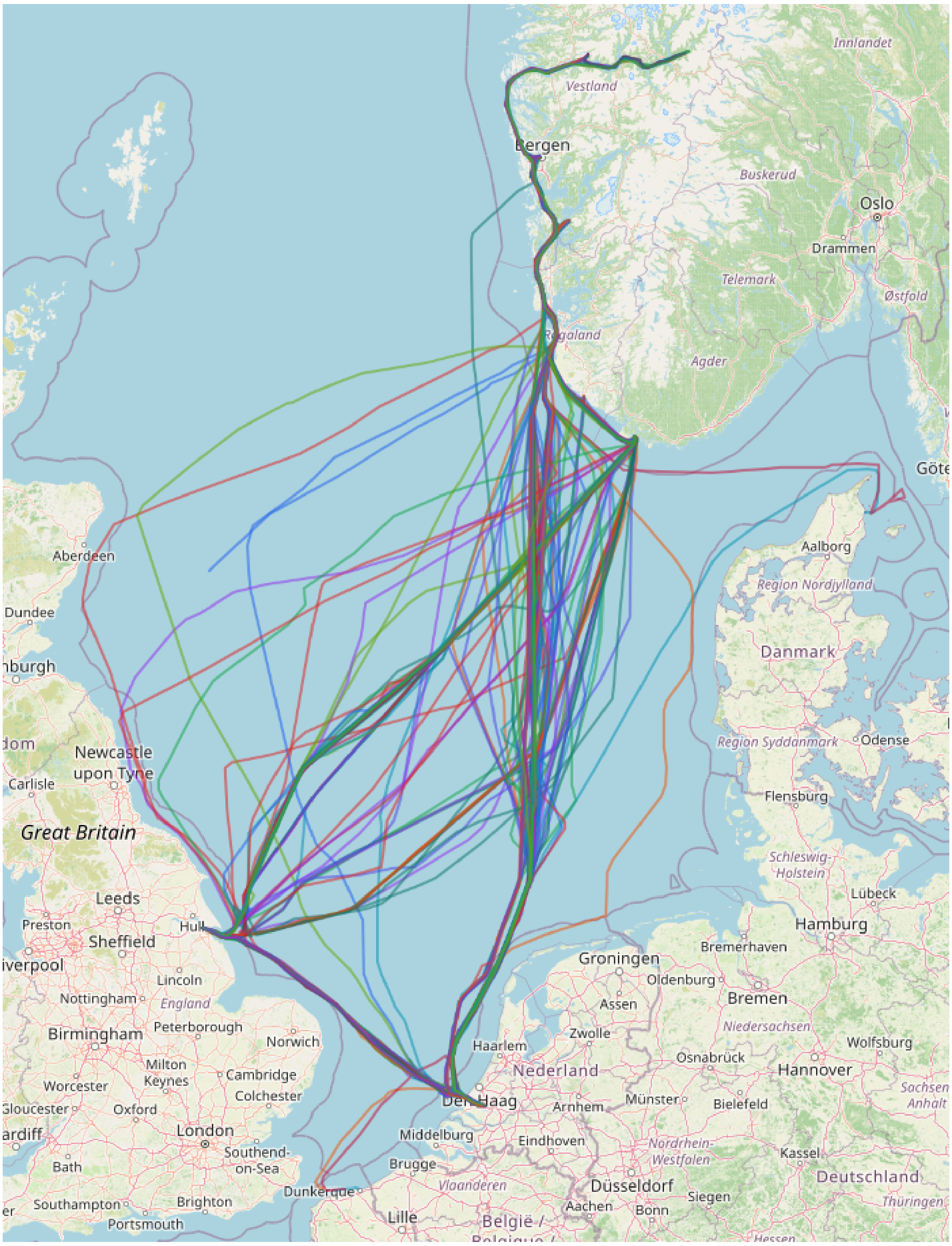


**Figure 1:** Training trajectories used by the project pipeline, shown on an OpenStreetMap basemap for geographic context [8].

## 3.3 Feature engineering

The engineered variables follow a leakage-aware design. Features are divided into explanatory metocean/route variables, lagged nowcast variables, and post-prediction rotor-scenario variables. The main feature groups are:

- **Wind and wave features**: true wind speed, significant wave height $H_s$, wave period, wind speed squared, and wave-wind alignment.
- **Relative-angle features**: wind direction and angle relative to vessel course, encoded with sine/cosine terms to avoid discontinuity at 0/360 degrees.

- **Apparent wind features**: apparent wind speed and apparent relative angle, reserved for the nowcast model.
- **Current features**: current magnitude, current direction, and along-course/cross-course current components.
- **Lagged target features**: previous SOG, two-step previous SOG, and a shifted rolling mean over up to three previous observations.
- **Interaction terms**: wind-angle, apparent-wind-angle, wind-wave, nonlinear wind/wave terms, and current-course interactions.

The most important leakage control is that the shifted rolling mean never uses the current target row. The lag features are grouped by voyage, so the first observations in a voyage do not inherit speed information from a previous voyage. This makes the nowcast model realistic for short-horizon prediction while keeping the weather-only model interpretable.

Angle handling deserves separate attention. Directions wrap around at 360 degrees, so raw angles can be misleading for a machine-learning model. A wind direction of 359 degrees and 1 degree are physically close, but numerically far apart if used as plain numbers. Sine and cosine encodings solve this by placing directions on a circle. The same idea is used for relative wind angle features, which are central to rotor performance.

Apparent wind features are also meaningful because the rotor experiences the wind relative to the moving vessel, not only the true wind over the sea. If a ship moves quickly into the wind, the apparent wind speed and angle can differ substantially from the true wind. The apparent-wind features therefore make the nowcast model closer to the physical situation on board. They are kept out of the simpler weather-only baseline so that the explanatory model remains closer to forecast-available variables and route context.

Current features are decomposed into along-course and cross-course components. This is more informative than using only current speed. A 0.5 m/s current helping the ship along its course has a different operational meaning from the same current acting sideways. Along-course current can directly affect observed SOG, while cross-course current may affect steering, track-keeping, and route deviation. Encoding the vector relative to course lets the model see this difference.

The lagged speed features are powerful but must be interpreted carefully. They capture persistence in vessel behaviour: a ship moving at 12 kn a few minutes ago is likely to remain near that speed unless there is a manoeuvre or a change in conditions. This is useful for nowcasting, but it can hide environmental effects. For example, if bad weather has already forced the ship to slow down, the lagged speed carries that consequence forward. That is why the report keeps separate weather-only and lagged-nowcast results.

**Table 3:** Representative variables in the engineered feature table.

| Group | Examples and role |
|---|---|
| Trajectory | `Time`, `Lat`, `Lon`, `Voyage_ID`, `Course_deg`, `Dist_since_last_nm`. |
| Target | `Speed_kn`; the predicted speed over ground. |
| Wind/wave | `True_Wind_Speed_ms`, `H_s`, `mwp`, `fg10`, wave-wind alignment. |
| Angles | `alpha_sin`, `alpha_cos`, apparent wind angle encodings. |
| Currents | `Current_Speed_ms`, along-course and cross-course current terms. |
| Nowcast history | `Speed_kn_lag1`, `Speed_kn_lag2`, `Speed_kn_mean3`. |
| Rotor scenario | Rotor power, active flag, and ΔSOG are used after model prediction, not as training features. |

# 4 Methodology

## 4.1 Speed prediction problem

The primary supervised learning target is vessel speed over ground:

$$\widehat{SOG}_t = f(X_t),$$

where $X_t$ contains environmental variables, route context, and, for the nowcast model only, previous observed speed. The models are evaluated on later voyages that were not used for training. This is essential because random row splits would allow neighbouring points from the same voyage to appear in both train and test sets, giving overly optimistic performance.

The project uses three conceptual model families:

1. **B1 constant mean**: predicts the mean training speed for every test observation.
2. **B2 linear weather model**: predicts speed from true wind speed, $H_s$, and sine/cosine encoding of the relative wind angle.
3. **XGBoost models**: gradient-boosted trees trained in two modes: weather-only explanatory modelling and lagged nowcast prediction.

The XGBoost implementation uses 300 estimators, maximum depth 6, learning rate 0.05, 80% row subsampling, 80% column subsampling, and a fixed random seed. These values were selected as a fixed manual configuration for reproducible course-project modelling rather than by a formal Optuna or GridSearch hyperparameter search. Features are standardised before model fitting in the implementation.

The model hierarchy is intentionally simple at the start. A constant mean model is not expected to be good, but it provides a useful reference point. If a more complex model cannot beat the average-speed baseline, then it has not extracted enough useful structure from the inputs. The linear model is the next step: it tests whether the core project variables of wind, wave height, and relative angle provide a clear first-order signal. XGBoost is then used because tree ensembles can capture nonlinear thresholds and interactions, which are expected in maritime performance data [7].

For example, wave height may have little effect below a moderate threshold and a much stronger effect above it. Wind can help or hurt depending on direction. Current can matter differently depending on course. These are interaction-heavy relationships, so a nonlinear model is a reasonable choice. At the same time, the baselines remain important because they prevent the project from claiming complexity as a result by itself.

## 4.2 Weather-only versus nowcast modelling

The report distinguishes two purposes that are easy to confuse:

- The **weather-only model** is an explanatory model. It excludes target-history features, so its behaviour is more directly related to wind, waves, currents, and route context.
- The **lagged nowcast model** is a predictive operational model. It uses previous speed observations and apparent wind context, making it much more accurate for short-horizon prediction.

This separation matters for interpretation. A high-performing nowcast model can mostly learn persistence: ships tend to continue moving near their recent speed. That is useful operationally, but it does not by itself prove that weather features are strongly explanatory. The report therefore presents both model types and treats weather interpretation with caution.

In practice, the two model types answer different user questions. The weather-only model asks: if we know the route context and forecast-like environmental variables, how much of speed can we explain? The nowcast model asks: if we also know how the vessel has just been moving, how accurately can we predict the next observed speed? Both are valuable, but mixing them would make the interpretation weaker. The weather-only model is closer to planning and analysis; the nowcast model is closer to monitoring and short-horizon prediction.

This distinction also affects feature importance. In the nowcast model, lagged speed naturally dominates. That does not mean wind and waves are irrelevant. It means recent speed is an efficient summary of many recent influences, including operator decisions and environmental resistance. The weather-only model is less accurate, but it gives a cleaner view of which forecast-available variables carry signal when speed history is not allowed to do most of the work.

## 4.3 Rotor scenario model

The rotor scenario is applied after baseline speed prediction. For each AIS observation, the rotor utility estimates a propulsion power $P_{\text{rotor}}$ from wind speed and relative wind angle using the supplied polar-response table. The rotor is active if:

$$P_{\text{rotor}} > 0, \quad H_s \leq 6 \text{ m}, \quad U_{\text{wind}} \leq 42 \text{ m/s}.$$

The speed contribution is then approximated as:

$$\Delta SOG = \frac{P_{\text{rotor}}}{200\text{kW/kn}}$$

when the rotor is active, otherwise $\Delta SOG = 0$. This is a transparent surrogate assumption rather than a full hydrodynamic propulsion model. Its value is that it makes the scenario auditable: the rotor benefit can be traced back to the wind-speed/angle polar, wave-height gate, and a single power-to-speed conversion.

The rotor model is best understood as a controlled scenario rather than a direct measurement. The historical AIS data represent actual vessel behaviour, and the vessel may already have used its installed rotors whenever conditions allowed. They therefore do not provide a pure no-rotor counterfactual or a clean experimental switch that tells us exactly when the rotor was off and on under identical conditions. The project builds a baseline prediction from observed operations and then applies a consistent rotor-opportunity layer derived from the supplied polar relationship. This creates a repeatable baseline-plus-scenario comparison for the same rows.

The conversion from rotor power to speed increment is deliberately simple. It does not replace a resistance and propulsion simulation. Instead, it provides an interpretable bridge between aerodynamic assistance and the speed-based questions in the project. Because the assumption is visible, it can later be replaced by a more detailed vessel model if engine curves, propeller efficiency, or measured rotor telemetry become available.

A specific limitation is that this section applies a linear speed-offset surrogate, while the later energy extension converts speed and power through a cubic resistance relationship. The two calculations are therefore consistent as transparent scenario approximations, but they should not be read as a single fully coupled hydrodynamic model.

## 4.4 Weather-window definition

The main weather-window criterion follows the project specification:

$$SOG \geq 10\text{kn} \quad \text{and} \quad H_s \leq 3 \text{ m}.$$

For every test-set observation the pipeline compares a baseline prediction with the same prediction after the rotor scenario layer is added. Contiguous observations satisfying the criterion form weather windows. The report tracks both the number of windows and the number/share of observations inside windows. Counting both is important: the scenario can reduce the number of separate windows if neighbouring windows merge into longer continuous periods.

The chosen weather-window definition is not the only possible operational definition, but it is easy to communicate. It combines a performance threshold with a sea-state threshold. The speed threshold asks whether the ship can keep a useful operating speed. The wave threshold asks whether the sea state remains within an acceptable range. The same framework can later be reused with stricter or looser thresholds depending on the operational question.

For this project, the window analysis is more informative than a single mean SOG value. A mean speed can hide whether gains happen during already favourable periods or during marginal periods. Weather windows show whether the scenario changes the classification of observations from unusable to usable. That is the kind of result that can matter for schedule planning and route choice.

### 4.5 Uncertainty evaluation

The baseline notebook also trains quantile-regression prediction intervals for the nowcast speed model. The reported 95% interval has mean lower bound 9.670 kn, mean upper bound 12.831 kn, mean width 3.161 kn, and empirical coverage 96.5% on the test set. This suggests that the interval model is slightly conservative but well calibrated for the current test distribution.

The interval result is useful because a deterministic point prediction can look more certain than it really is. In operational planning, uncertainty affects risk: a predicted speed of 10.1 kn is not equally useful if the uncertainty range is narrow or if it spans several knots. Prediction intervals give a way to communicate this risk. In later versions, the same idea could be applied directly to weather-window decisions by estimating the probability that a row or segment satisfies the threshold.

## 5 Model results and evaluation

### 5.1 Dataset split

The train/test split is chronological by voyage. The first 80% of voyages by start time are used for training; the final 20% are used for testing. The baseline notebook reports:

- Training set: 88,346 rows from 422 voyages.
- Test set: 35,758 rows from 106 voyages.
- Training period: 2023-02-02 to 2025-05-23.
- Test period: 2025-05-23 to 2026-02-02.
- Temporal leakage check: true.

This split is stricter than a random split and is therefore a stronger estimate of generalisation to future voyages.

The chronological split also makes the reported results easier to defend. In trajectory data, neighbouring rows are highly correlated. Randomly assigning rows to train and test would allow the model to learn from points immediately before or after a test point, which is not a realistic future-voyage setting. Splitting by voyage reduces that risk. Splitting chronologically adds another layer of realism, because the test set is later in time than the training set.

This does make the modelling task harder. Seasonal patterns, changing routes, maintenance state, or operating policy can differ between early and late voyages. That is precisely why the split is useful. It tests whether the model can carry learned relationships into later data, not merely interpolate inside a single voyage.

### 5.2 Regression metrics

The model comparison is summarised in Table 4. The constant mean and linear weather model are close to one another. The weather-only XGBoost model performs poorly relative to the linear baseline on the held-out test period, which suggests either distribution shift, missing operational variables, or that a large part of speed variation is driven by control decisions and vessel state not visible in the current feature table. The lagged nowcast model is much stronger because recent observed speed is a powerful predictor.

MAE, RMSE, and $R^2$ each tell a different part of the story. MAE is easy to interpret because it is the average absolute error in knots. RMSE penalises larger mistakes more heavily, so it is sensitive to occasional poor predictions. $R^2$ describes explained variance relative to a mean baseline. The within-$\pm1$ kn metric is included because it is operationally intuitive: it tells how often the prediction is close enough to be practically useful.

The gap between M1 and M2 is one of the most important evaluation results. The weather-only model does not generalise strongly on the held-out voyages, while the lagged nowcast model performs well. This suggests that the current dataset contains enough information for short-horizon prediction, but that weather-only speed prediction is limited by missing operational variables and by the complexity of vessel behaviour.

There is also a useful modelling paradox in the comparison between B2 and M1: the linear weather baseline reaches $R^2 = 0.034$, while the weather-only XGBoost model reaches only $R^2 = 0.010$ on the same chronological test period. Because a linear model outperforms a more flexible gradient-boosted tree on this feature set, the likely explanation is not that the relationship is truly linear, but that the XGBoost model has fitted training-period structure that does not transfer to later voyages. In future iterations this should be tested with stricter regularisation, for example larger $L_1/L_2$ penalties (`reg_alpha`, `reg_lambda`), a larger `min_child_weight`, shallower trees, and validation by rolling or nested chronological splits.

**Table 4:** Held-out test performance for the core model hierarchy.

| **Model** | **MAE** [kn] | **RMSE** [kn] | $R^2$ | **Within** ±1 kn |
|---|---|---|---|---|
| B1: Constant mean | 1.388 | 1.893 | -0.002 | 49.3% |
| B2: Linear wind/wave/angle | 1.377 | 1.860 | 0.034 | 49.5% |
| M1: XGBoost weather-only | 1.468 | 1.883 | 0.010 | 41.7% |
| M2: XGBoost lagged nowcast | 0.406 | 0.758 | 0.839 | 91.7% |

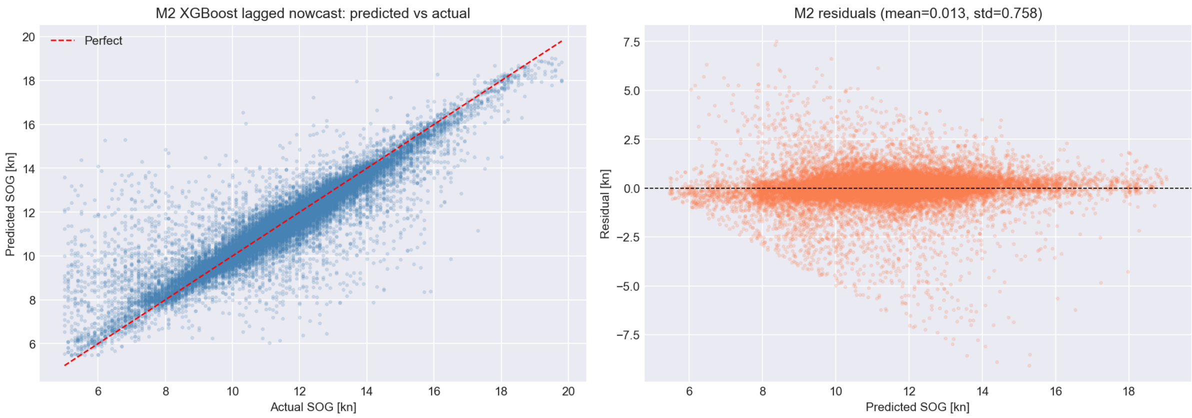


**Figure 2:** Predicted versus actual SOG for the lagged nowcast model.

## 5.3 Feature importance

The weather-only model ranks course and current interaction terms highly, while the lagged nowcast model is dominated by recent observed speed. This is consistent with vessel operations: the current speed carries information about engine setting, traffic, manoeuvring, loading, and other operational factors that are not explicitly modelled.

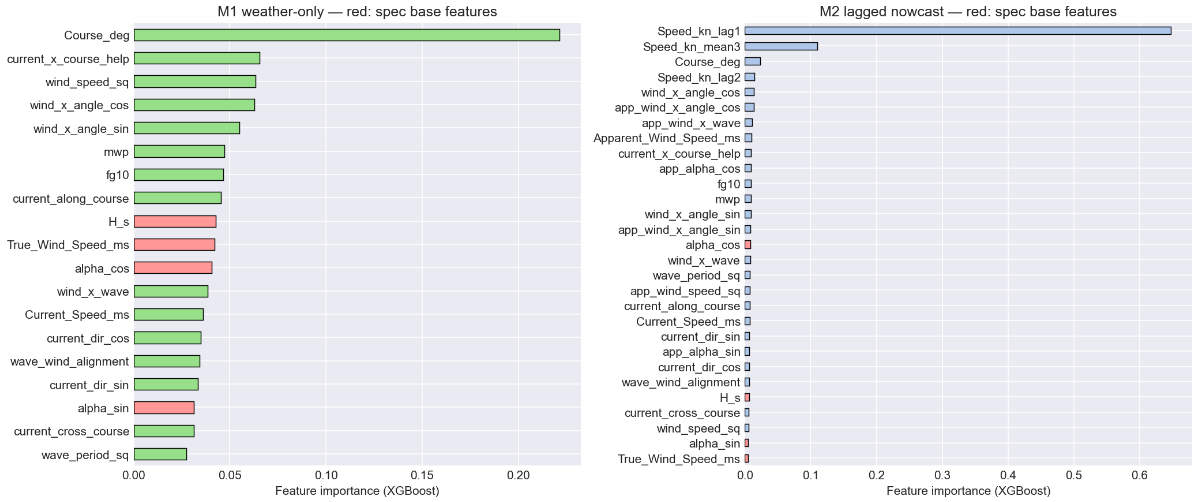


**Figure 3:** Feature importances for weather-only and lagged nowcast XGBoost models.

The main interpretation is therefore twofold. First, the project can now predict short-term SOG accurately when recent speed is available. Second, pure weather-to-speed inference is harder than expected and should be treated as an explanatory sensitivity tool rather than a final operational predictor.

The feature-importance plot should not be read as a strict physical ranking. Tree-based importance depends on the model, the correlated features, and the training distribution. Still, it is useful for checking whether the model is behaving plausibly. In the weather-only case, route and current-related features appear alongside wind and wave variables. In the lagged model, recent speed dominates, which is exactly what we would expect from a nowcast formulation.

One practical interpretation is that future work should not rely only on more complex models. Additional explanatory variables may matter more than adding model complexity. Draught, engine power or RPM, loading condition, port approach flags, and a reliable rotor-operation label could all improve the ability to separate weather effects from operating decisions.

# 6 Rotor and weather-window results

## 6.1 Rotor what-if scenario

On the test set, the rotor surrogate is active for 35,077 of 35,758 rows, or 98.1% of observations. The mean rotor power when active is 41.2 kW. The mean active speed increment is 0.206 kn, and the overall mean increment is 0.202 kn. Per-voyage results are heterogeneous: the highest-gain voyages in the test set show mean speed contributions from about 0.47 kn to 1.19 kn, while the overall mean remains modest because many observations have weaker wind, less favourable angle, or low rotor power.

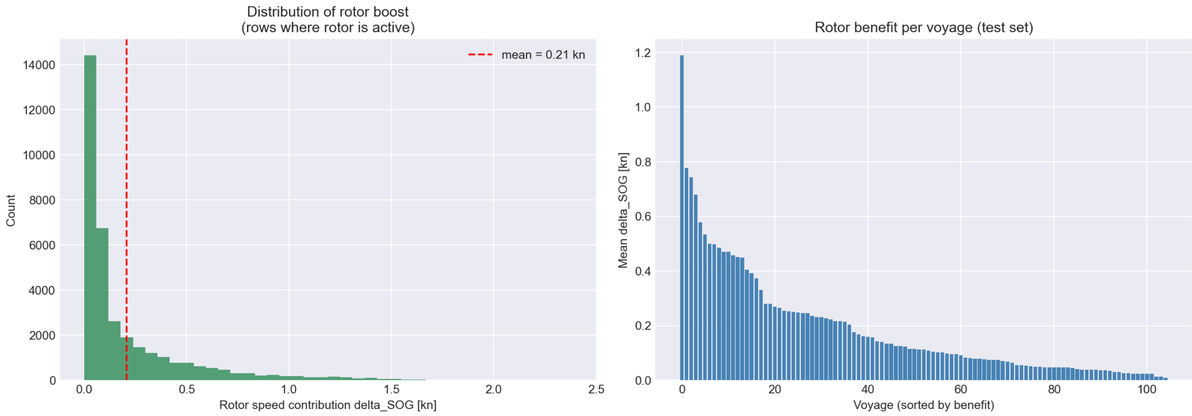


**Figure 4:** Rotor scenario distribution and per-voyage mean speed contribution.

The result is operationally meaningful but not spectacular in average speed terms. An average 0.2 kn speed uplift can still matter over long voyages, especially if it moves marginal periods over a speed threshold or allows lower engine load for the same schedule. The larger gains occur in a smaller subset of conditions, which motivates the gain-detection extension.

The rotor scenario also shows why averages can be misleading. A mean gain of 0.2 kn sounds small, but it is averaged over many different wind angles and wind speeds. The per-voyage table and gain-detection analysis show that some voyages experience much larger contributions. For a decision-support tool, this means the useful question is not only "what is the average gain?", but also "where are the high-gain periods and can we plan around them?"

The active share is high because the implemented gate is permissive: rotor power only needs to be positive, wave height must be below 6 m, and wind speed must be below the stop threshold. Active does not mean high impact. It only means the scenario allows a positive contribution. The high-gain classifier later adds a more selective view by focusing on observations where the modelled speed contribution is at least 0.5 kn.

## 6.2 Weather-window impact

The weather-window analysis evaluates whether the rotor scenario helps maintain the criterion SOG $\geq 10$ kn and $H_s \leq 3$ m. The result is:

**Table 5:** Weather-window comparison on the held-out test set.

| Scenario | Windows | Points in window | Coverage |
|---|---|---|---|
| Baseline prediction | 866 | 27,167 | 76.0% |
| Baseline + rotor scenario | 808 | 27,696 | 77.5% |
| Delta | -58 | +529 | +1.5 percentage points |

The decrease in the number of windows should not be read as a negative result. When the rotor scenario lifts speed above the threshold between two favourable periods, separate windows can merge. The more relevant operational result is that the number of points inside the window increases and total coverage rises.

The window result is best interpreted as a marginal improvement in operational availability. Coverage rises by 1.5 percentage points on the test set. That is not a transformation of the whole operating profile, but it is a measurable change. For a vessel that repeats similar routes many times, small changes in coverage can accumulate into fewer delays, more schedule flexibility, or lower fuel use for the same arrival target.

The reduction in the number of windows is a good example of why the project tracks several indicators. If only the count of windows were reported, the rotor scenario would look worse. Adding the number of points and coverage shows the opposite: favourable periods become more continuous. This is exactly the kind of interpretation that matters when translating model output into maritime operations.

### 6.3 High-gain geography and environmental structure

The gain-surface and interaction plots show that rotor benefits cluster in particular wind-speed and angle regimes. This is expected from the polar diagram: headwind and tailwind directions are poor, while beam and broad-reaching angles are stronger. Wave height mostly acts as a boundary or risk condition; in the implemented scenario the rotor is forced off above $H_s = 6$ m, and only 19 observations exceed that threshold in the full engineered dataset.

The gain surface provides the clearest operational message in the rotor analysis. Rotor benefit is concentrated in a band of wind directions rather than spread uniformly across all directions. This matches the physical expectation from the Magnus effect: a rotating cylinder produces a force roughly perpendicular to the apparent wind, and the useful part of that force depends on how it projects onto the vessel's direction of travel.

The wave-height interaction is different. Wave height is not the main source of positive rotor power in this scenario, but it controls whether the scenario is considered operationally acceptable. In other words, wind angle and speed create the opportunity, while sea state can remove or limit that opportunity. This distinction is useful for later route planning because it suggests that a good rotor route must be both windy in the right way and safe enough in waves.

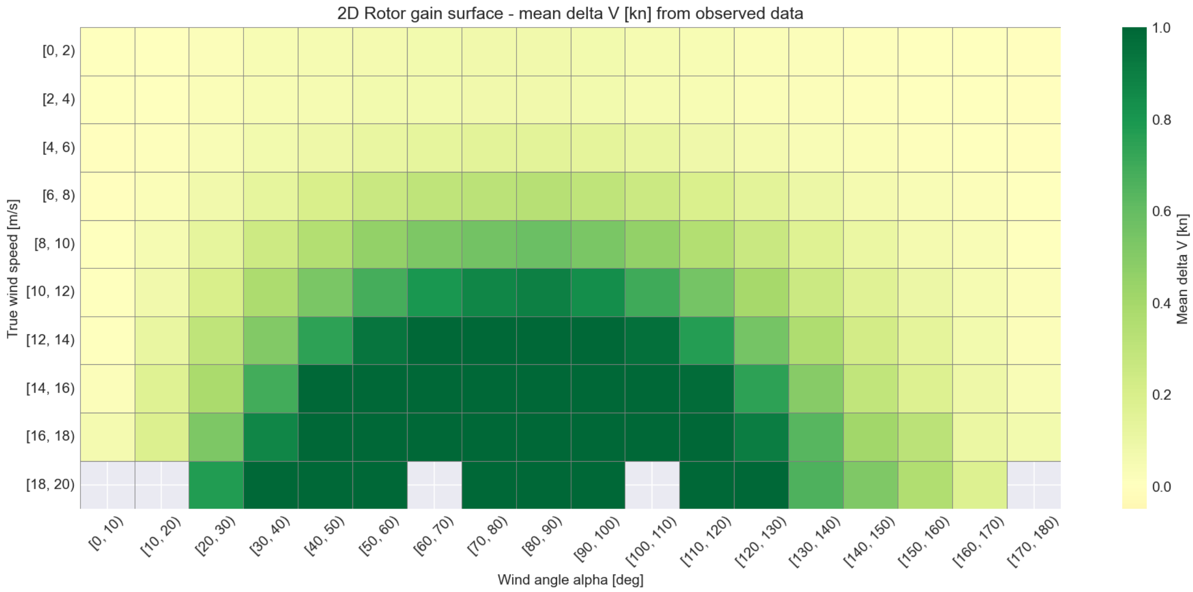


**Figure 5:** Environmental structure of the rotor-gain scenario.

## 7 Extensions to the project

### 7.1 Energy, emissions, and CII extension

The first extension converts rotor power into an energy and emissions impact. The ship model assumes 3,500 kW at design speed 11.5 kn, a cubic resistance relationship, specific fuel oil consumption of 195 g/kWh, and an HFO $CO_2$ factor of 3.114 t $CO_2$/t fuel. Under these assumptions the fuel rate at design speed without the rotor is 16.38 t/day.

The three-year dataset summary is:

- Total observations: 125,160.
- Rotor active share: 97.6%.

- Mean brake power: 3,887 kW.
- Mean rotor power when active: 40.4 kW.
- Total fuel, baseline estimate: 11,066.7 t.
- Total fuel, rotor-scenario estimate: 10,954.4 t.
- Fuel saved: 112.2 t, or 1.01%.
- $CO_2$ saved: 349.4 t, or 1.01%.

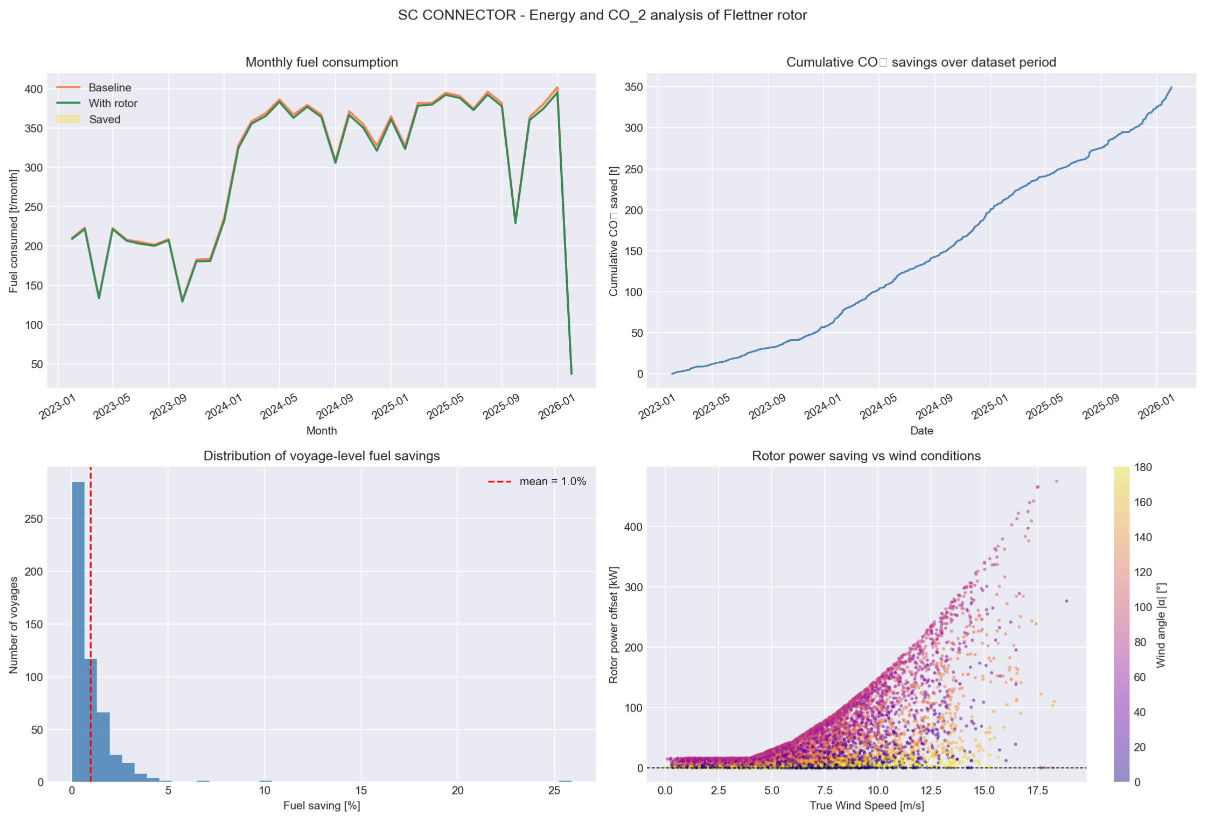


**Figure 6:** Fuel and $CO_2$ impact estimated from the rotor energy model.

Annualised over the dataset duration of 1,096 days, the extension estimates 37.4 t fuel saved per year and 116.4 t $CO_2$ avoided per year. The attained CII estimate improves from 53.75 to 53.20 g $CO_2$/(t nm), a 1.01% improvement.

The conclusion is that the average annual energy impact is modest for one rotor, but it is measurable and directionally useful. The result should be interpreted as an order-of-magnitude scenario rather than a guaranteed vessel-performance claim, because the historical data do not provide a clean no-rotor control and the model does not yet include detailed propeller curves, engine load-specific SFOC, hull fouling, loading condition, RPM control policy, or captain/charterer speed decisions.

The energy extension is valuable because it translates a speed-oriented rotor scenario into fuel and emissions language. A small speed increment can be used in two ways. The operator can keep the same engine setting and arrive slightly earlier, or reduce engine load and keep approximately the same schedule. The notebook frames the result as a fuel-saving scenario, which is often the more important business and environmental interpretation.

The cubic resistance assumption is a standard simplification: required power increases rapidly with speed. This means that modest speed changes can correspond to meaningful power differences, especially around normal operating speed. However, the simplification also means the absolute fuel numbers should be treated as scenario estimates. The most robust finding is the direction and order of magnitude of the saving, not an exact guarantee for every voyage.

CII is included because shipping decarbonisation is not only a technical issue but also a regulatory and reporting issue. Even a roughly one-percent improvement can be relevant when a vessel is close to a rating boundary or when operators combine several efficiency measures. The rotor alone is not presented as a complete decarbonisation solution. It is one operational lever that can contribute alongside routing, speed management, hull maintenance, and fuel choice.

## 7.2 Route segmentation extension

Route segmentation divides the data into heading regimes: northbound, eastbound, southbound, and westbound. This makes the analysis more interpretable because rotor performance is strongly tied to the relationship between course and wind. A single global model can hide differences between legs moving through different prevailing-wind relationships.

**Table 6:** Regime-level characterisation from the full dataset.

| **Regime** | **Mean SOG** [kn] | **Mean wind** [m/s] | **Mean** $H_s$ **[m]** | **Mean** $\Delta$**SOG** [kn] |
|---|---|---|---|---|
| Northbound | 12.05 | 7.0 | 1.32 | 0.209 |
| Eastbound | 12.34 | 6.2 | 1.06 | 0.185 |
| Southbound | 10.95 | 6.8 | 1.28 | 0.196 |
| Westbound | 10.58 | 6.5 | 1.21 | 0.209 |

The per-regime weather-only XGBoost models show mixed predictive quality. Northbound has the lowest MAE at 1.219 kn, while westbound is hardest to predict at MAE 1.788 kn. Eastbound is the only regime with clearly positive $R^2$ in the per-regime model summary, $R^2 = 0.137$, while the other regimes have negative $R^2$. This again supports the conclusion that weather-only generalisation is difficult.

The most useful output from this extension is the weather-window effect by heading. Westbound legs improve from 85.1% to 90.1% coverage, the largest regime-level gain. Southbound improves from 82.3% to 84.5%. Northbound and eastbound already have high baseline coverage and therefore show smaller coverage changes.

The regime results show that the same rotor can have different value depending on route direction. This is expected because a vessel changing heading changes the relative wind angle even under the same weather system. A northbound leg and a westbound leg can therefore experience different rotor geometry without any change in the rotor itself.

The segmentation also helps interpret prediction difficulty. Some regimes are more regular than others. A route direction with consistent operating patterns and environmental exposure may be easier to model than a regime that includes more manoeuvring, port approaches, or mixed conditions. The negative $R^2$ values in several per-regime weather-only models are a warning that splitting the data does not automatically solve the problem. It improves interpretability, but each segment still needs enough data and enough explanatory variables.

From an operational perspective, the westbound result is especially interesting because weather-window coverage changes more strongly there than in the other regimes. That does not mean westbound is always best for rotors. It means that, in this dataset and under the chosen threshold, westbound observations contain more marginal cases that the rotor scenario can move into the acceptable window.

## 7.3 Route optimisation extension

The route-optimisation extension explores whether route choice can increase the expected rotor benefit. The implemented proof of concept builds a coarse 0.5-degree spatial grid from historical metocean/rotor-gain fields and applies Dijkstra search to identify a path with improved integrated rotor gain. The reference voyage is Voyage 456, from approximately 51.89 degrees N, 4.40 degrees E to 61.23 degrees N, 7.68 degrees E. The great-circle distance is about 570.7 nm.

The spatial field contains 72 grid cells with data. Mean rotor power across the grid is 78.5 kW, and the best historical cells reach mean rotor powers above 180–290 kW under favourable conditions. The route-search result reports an optimal path of 19 nodes, distance 568.0 nm versus 571.0 nm for the direct grid path, mean rotor power 7.5 kW versus 1.9 kW direct, and mean ΔSOG 0.037 kn versus 0.010 kn direct. Relative improvement is large at +288.5%, but the absolute gain in this specific route experiment is small.

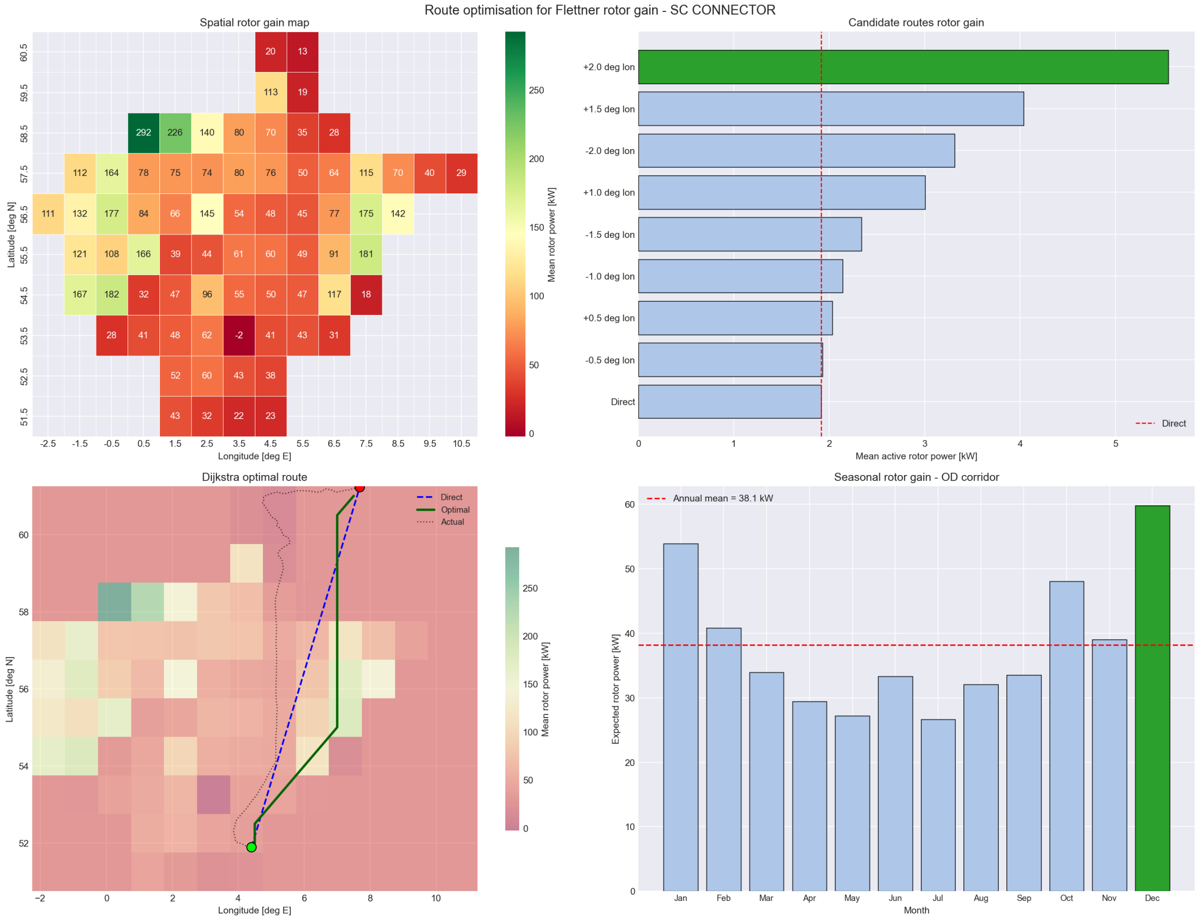


**Figure 7:** Dijkstra-based routing proof of concept for rotor-gain optimisation.

The extension also studies seasonal sensitivity. December has the best expected monthly rotor power, 59.8 kW, while July is worst at 26.6 kW. The seasonal pattern is intuitive for the North Sea: winter has stronger winds, but it may also have more severe waves and operational risk. A final routing product should therefore optimise a multi-objective score, not rotor gain alone. A practical score would combine expected arrival time, fuel, $CO_2$, wave limits, route deviation, safety margin, and schedule reliability.

The routing extension should be read as a proof of method rather than a final voyage planner. The current grid is coarse and built from historical fields. That is enough to show that route geometry can be connected to rotor gain, but a production routing tool would need forecast fields, land and shallow-water masks, traffic separation schemes, port constraints, and safety rules.

Dijkstra search is appropriate for a first version because it makes the route choice explicit. Each grid node and edge can be assigned a cost, and the algorithm finds the path with the lowest cumulative cost. In this project, the cost is related to rotor opportunity and route length. In a more complete version, the edge cost could combine fuel, time, wave risk, distance from the planned corridor, and uncertainty. The same framework could therefore evolve from a gain-maximisation example into a multi-objective decision tool.

The seasonal result is also important. The best month for rotor power is not automatically the best month for operations, because stronger wind can coincide with higher waves and stricter safety constraints. This reinforces the report's main theme: rotor benefit is conditional. A good decision-support tool should not simply search for the strongest wind; it should search for useful wind under acceptable sea-state and route constraints.

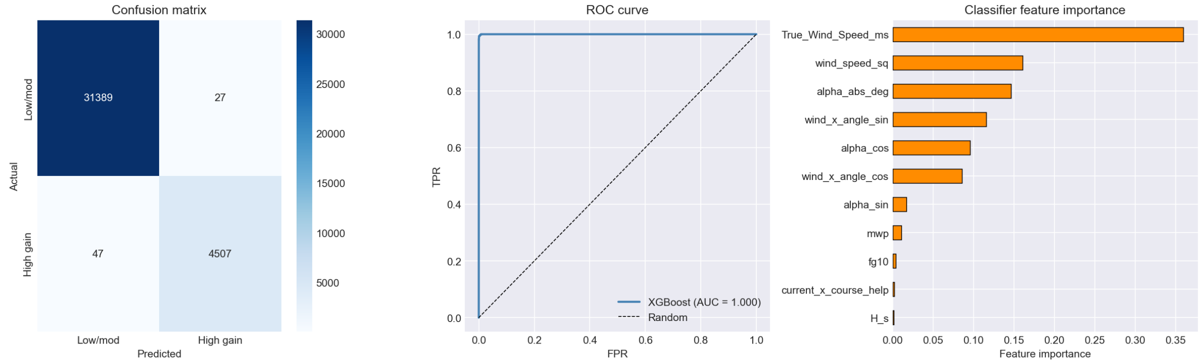


**Figure 9:** Classifier performance.

## 7.4 Rotor gain detection extension

The gain-detection extension turns the rotor scenario into a classification and decision problem. A high-gain observation is defined as $\Delta SOG \geq 0.5$ kn. Under the current surrogate model, high-gain moments account for 14,856 observations, or 11.9% of the full dataset.

Rotor gain increases strongly with wind speed. The notebook reports mean $\Delta SOG$ of 0.037 kn for wind below 4 m/s, 0.273 kn for 8–10 m/s, 0.431 kn for 10–12 m/s, 0.610 kn for 12–14 m/s, and 0.897 kn above 14 m/s. The optimal high-gain subset has median wind speed 11.5 m/s, and 90% of those observations are above 9.2 m/s.

Angle is equally important. High-gain observations have median absolute relative wind angle around 82 degrees, with 80% of the main high-gain subset between 50 and 113 degrees. This matches the rotor polar: beam and quartering winds are useful, while headwind and tailwind sectors are poor.

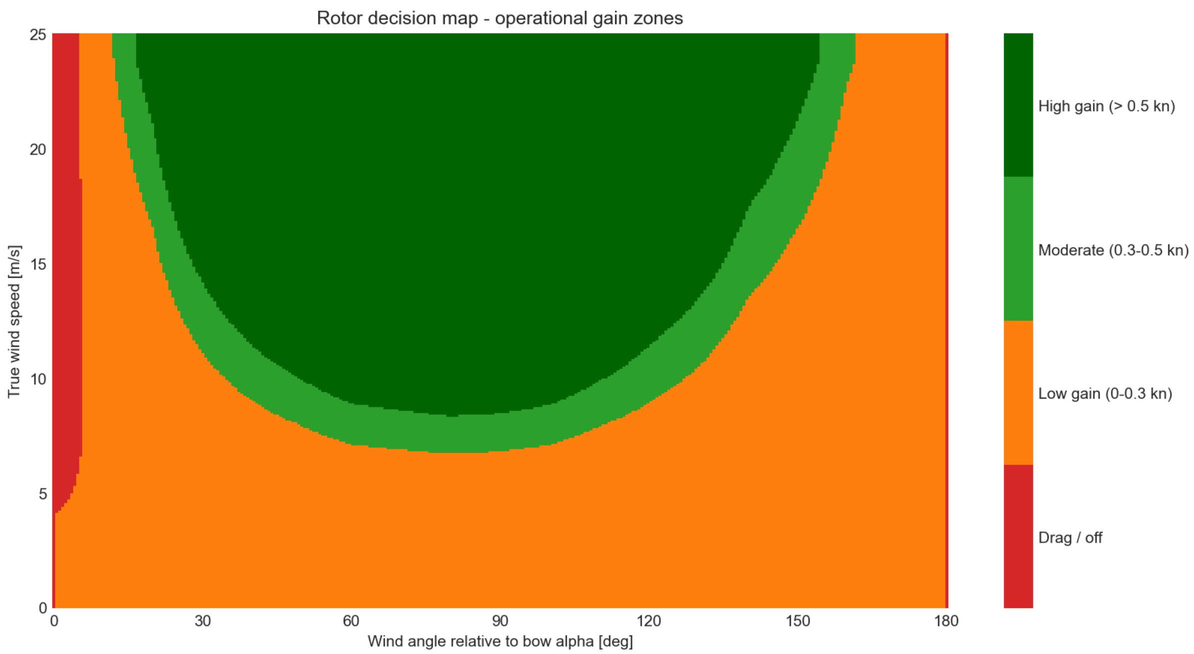


**Figure 8:** Rule-based high-gain zones.

An XGBoost classifier identifies high-gain moments with 99.8% accuracy, 99.4% precision, 99.0% recall, F1 = 0.992, and ROC-AUC = 1.000 on the notebook split. These very high metrics are plausible because the target is derived directly from the same rotor-power surrogate and its input variables; the classifier is therefore best interpreted as a compact decision-rule learner, not as independent proof of physical rotor performance.

The rule-based decision system is more transparent: if wind is at least 8 m/s and angle is between 50 and 130 degrees, or wind is at least 12 m/s and angle is between 40 and 140 degrees, mark high gain. This rule reaches 95.8% recall, 69.7% precision, and 94.2% accuracy. For operations, the rule is attractive because it is easy to explain and can be applied to forecasts.

The classifier and the rule-based system serve different purposes. The XGBoost classifier is accurate because it learns the structure of the model-derived high-gain label. It is useful as a compact approximation of the scenario model, but its high score should not be mistaken for independent physical validation. The transparent rule has lower precision but is easier to communicate: it says that meaningful rotor moments mostly occur above a wind-speed threshold and inside a broad favourable angle sector.

This kind of rule is useful for a bridge or operations dashboard. Instead of showing only a continuous power number, the dashboard can highlight periods as low, moderate, or high rotor opportunity. A planner can then compare those periods with route alternatives, schedule constraints, and wave limits. The result is more actionable than a single annual fuel-saving estimate because it points to the moments where operational attention matters most.

High-gain detection also helps avoid overclaiming. If only 11.9% of observations are high-gain moments, then the rotor's value is concentrated. This supports a nuanced conclusion: the rotor is not a universal speed boost, but it can be useful when the environmental geometry is right. That message is both technically honest and operationally practical.

## 7.5 Monte Carlo routing extension

The latest extension turns route optimisation into an uncertainty problem. The earlier routing notebook asks whether one deterministic path can improve rotor opportunity. The Monte Carlo notebook asks a stricter operational question: which route performs well across many plausible wind, wave, and current realisations, not only under one averaged condition?

The reference case is Voyage 364. The notebook first reconstructs the observed itinerary and separates it into sailing legs using stop gaps longer than two hours. This preserves the port-call structure of the voyage: optimisation is performed leg by leg between observed itinerary points, rather than by allowing the algorithm to replace the voyage with one unrealistic direct line. In the current run the method detects four sailing legs and three stop gaps.

The route area is converted into a navigable sea graph. Candidate nodes are placed on a coarse latitude-longitude grid with 0.35 degree spacing and kept only where they are close to historically observed AIS water corridors. Edges connect neighbouring grid cells in eight directions, with distance calculated by the haversine formula and course calculated from the start and end points. Several feasibility checks are then applied: edge midpoints must remain close to observed water traffic, diagonal moves cannot cut across unavailable orthogonal neighbours, and a Natural Earth land mask removes land-crossing edges. A small AIS-corridor override protects real near-port movements from being removed by an over-simple coastline mask. The current graph has 537 nodes, 3,784 edges, and 132 AIS-corridor override edges.

For every edge, the extension samples many plausible metocean scenarios from the historical feature table. This is empirical sampling rather than synthetic weather generation. Records are grouped into 0.7 degree spatial cells, and the preferred sample pool uses the voyage month plus or minus one month so that the draws remain seasonally realistic. If a local cell has too few observations, the sampler falls back to the nearest populated cell and then to the global seasonal sample. Each sampled scenario carries true wind speed, wind direction, significant wave height $H_s$, and the eastward and northward current components `uo` and `vo`. Sampling full historical records helps preserve realistic relationships between wind, waves, and currents.

Each edge is then evaluated under each sampled scenario. Current is projected onto the edge course, wave height applies a speed penalty above a calm-water threshold, and rotor power is estimated from the same polar-based scenario used elsewhere in the project. The notebook converts the resulting assisted speed and propulsion load into fuel and $CO_2$ using the same simplified energy model as the fuel extension. This produces a distribution of edge costs rather than a single deterministic edge cost.

The notebook compares five route concepts:

- Original observed route: the AIS path actually sailed, evaluated under the same scenario model.
- Shortest feasible route: graph route with minimum distance, ignoring weather uncertainty except through feasibility filters.

- Expected-cost Monte Carlo route: graph route minimising mean rotor-assisted fuel across the sampled scenarios.
- Risk-aware Monte Carlo route: graph route minimising mean fuel plus a penalty for scenario-to-scenario fuel variability.
- Scenario-optimal upper bound: an optimistic benchmark where each scenario is allowed to choose its own best route after the weather is known.

The upper bound is especially important to explain correctly. It is not a deployable route recommendation, because it assumes perfect knowledge of the realised weather for every sampled scenario. Its value is diagnostic: it shows how much extra benefit could theoretically exist if uncertainty were removed. The gap between the expected-cost route and the upper bound is therefore a measure of remaining weather-routing value, not a promise that the upper-bound result can be achieved operationally.

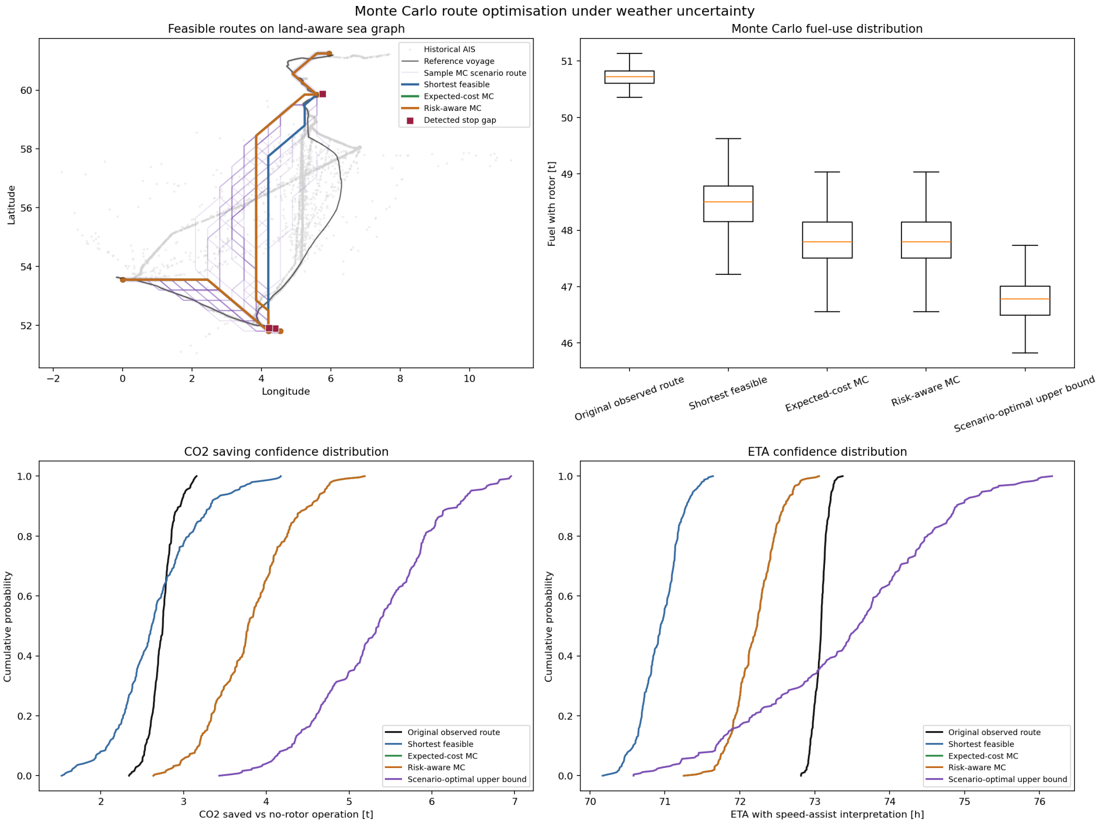


**Figure 10:** Monte Carlo route comparison for Voyage 364 under sampled metocean uncertainty.

**Table 7:** Monte Carlo route comparison for Voyage 364.

| Route | Distance [nm] | Mean fuel [t] | Fuel saved [t] | Beats shortest |
|---|---|---|---|---|
| Original observed | 863.57 | 50.72 | 0.88 | 0.0% |
| Shortest feasible | 834.07 | 48.47 | 0.86 | 0.0% |
| Expected-cost MC | 847.89 | 47.81 | 1.22 | 85.6% |
| Risk-aware MC | 847.89 | 47.81 | 1.22 | 85.6% |
| Scenario-optimal bound | 860.88 | 46.75 | 1.70 | 100.0% |

The result shows why uncertainty-aware routing is useful. The shortest feasible route is shorter than the Monte Carlo route, but it is not the best route under the sampled fuel objective. The expected-cost route accepts a small distance increase in exchange for more favourable assisted propulsion conditions. In the current run the risk-aware route equals the expected-cost route, which means the variability penalty was not strong enough to change the chosen path. That is not a failure; it tells us that, under the chosen risk parameter, the same route is both low-mean-cost and sufficiently stable.

The extension should still be presented as a decision-support prototype rather than a finished navigation system. The graph is coarse, the weather sampling is historical rather than a live ensemble forecast, and the land mask is not a nautical chart. Its value is methodological: it connects AIS corridors, route constraints, empirical metocean uncertainty, rotor physics, fuel, $CO_2$, ETA, and risk into one reproducible framework.

# 8 Practical interpretation

The practical interpretation of the project is that wind-assisted propulsion is best handled as an operational opportunity map. The rotor benefit depends on where the vessel is, where it is going, what the wind is doing, and whether sea state allows the route to remain useful. The project does not claim that one rotor changes every voyage in the same way. Instead, it identifies the conditions in which the effect is likely to matter.

For voyage planning, the most useful outputs are the weather-window comparison, the high-gain decision map, and the route-segmentation results. Together they answer three practical questions. First, does rotor assistance change the availability of acceptable operating periods? Second, what wind-speed and angle zones should the operator look for? Third, which route directions or voyage segments appear to benefit more under the current data?

For monitoring during a voyage, the lagged nowcast model is more relevant. It uses recent observed speed and apparent-wind context to predict near-term speed with much smaller error than weather-only models. That can support alerts when the vessel is likely to fall below a speed threshold, or when actual performance differs from what the model expects under the observed conditions.

For emissions reporting, the energy extension provides a bridge from meteorological opportunity to fuel and $CO_2$ estimates. The exact values depend on assumptions, but the workflow shows how a rotor scenario can be translated into quantities that are meaningful for owners, operators, and regulators. This is important because technical performance is only part of the adoption question; the business case usually depends on fuel cost, emissions reduction, schedule reliability, and operational simplicity.

# 9 Limitations and next steps

The main limitation is that the dataset does not fully describe vessel operation. AIS and metocean data are enough to model observed speed patterns, but they do not reveal all causes. Engine settings, loading condition, draught, propeller state, hull fouling, traffic, port manoeuvres, and operator decisions can all affect SOG. Without those variables, the weather-only model has a limited ceiling.

The rotor scenario is another limitation. It is based on the supplied polar diagram and a transparent power-to-speed conversion, not on measured onboard rotor telemetry. Because the historical period may already include rotor use, the baseline is not a certified no-rotor counterfactual. This is acceptable for a what-if prototype, but future work should validate the scenario against measured rotor power, shaft power, fuel flow, or controlled before/after periods if such data become available.

The route-optimisation extension should also be expanded carefully. Optimising only for rotor gain can create routes that are not operationally optimal. A realistic route score should include distance, time, safety, wave exposure, forecast uncertainty, navigational constraints, and schedule commitments. The current implementation is valuable because it demonstrates the graph-based method, but the cost function should become richer before operational use.

The most useful next steps are to export a consolidated results CSV, add segment-level travel-time error, test several weather-window thresholds, and run sensitivity checks on the rotor power-to-speed conversion. These additions would make the project easier to review and would show how stable the main conclusions are under reasonable changes in assumptions.

# 10 Conclusions

The project has progressed beyond a simple baseline and now forms a coherent maritime analytics prototype. AIS trajectories are enriched with wind, waves, and currents; leakage-aware models are trained and evaluated by voyage; rotor assistance is simulated as a transparent what-if layer; and the results are translated into weather windows, fuel, emissions, route regimes, routing, and high-gain operational rules. The Monte Carlo extension adds the final missing decision-support layer by showing how routing choices change when weather and current uncertainty are represented as a distribution instead of a single average condition.

The central technical finding is that recent vessel speed is the strongest predictor of near-term speed. The lagged nowcast model reaches MAE 0.406 kn and $R^2 = 0.839$, while weather-only models remain much weaker. This distinction should be presented honestly: the project can predict speed well in a nowcast setting, but causal weather-only interpretation is still limited by missing operational variables.

The central rotor finding is that the average benefit of one rotor is moderate but operationally relevant. Mean speed uplift is about 0.2 kn across the test set, weather-window coverage increases by 1.5 percentage points, and the energy extension gives a scenario estimate of about 37 t/year fuel saving and 116 t/year $CO_2$ reduction. The most useful message is conditionality: the rotor is most valuable in a minority of high-gain moments, especially when wind is above roughly 8–12 m/s and the apparent wind angle is near beam/quartering sectors.

The Monte Carlo result reinforces the same message at route level. In Voyage 364, the expected-cost route is not simply the shortest route; it is the route with the best mean fuel performance under sampled wind, wave, and current conditions. The scenario-optimal bound shows that additional routing value may exist if forecast certainty improves, while the gap to that bound keeps the claim realistic.

# 11 Transparency statement

Generative AI tools were used for language polishing, paraphrasing, and summarisation. The authors performed the analysis, made the modelling and interpretation decisions, and reviewed the final text for accuracy and relevance. The tools did not have access to the raw data and did not make modelling decisions.

# A Implementation notes

The repository is organised as a reproducible notebook pipeline. The earlier notebooks prepare and inspect the merged data; the feature-engineering notebook creates target-safe variables; the baseline model notebook trains and evaluates the main speed models; the extension notebooks reuse the same feature table for energy, segmentation, routing, gain detection, and Monte Carlo uncertainty routing.

The most important implementation choice is that rotor variables are not used as ordinary model features for the main speed prediction. They are applied after prediction as a scenario layer. This keeps the baseline model and the rotor-assisted scenario conceptually separate. If rotor contribution were mixed directly into the training features without a reliable ON/OFF label, it would be harder to explain what the model had actually learned.

The notebooks also export figures to the `results/` directory, which allows the report to reference a stable set of outputs. This makes the report less dependent on notebook display state and easier to regenerate. A useful next engineering step would be to export the most important numerical summaries to a single compact CSV, so that the report, presentation, and any reviewer checks all draw from the same result table.

## A.1 Assumptions kept consistent

The analysis depends on several simplifying assumptions. The most important point is that these assumptions are used consistently across the notebooks rather than changed from one extension to another. The rotor polar relationship is the same scenario utility used for speed uplift, fuel estimation, route opportunity, and high-gain classification. The voyage-based split is also kept as the validation principle for the main predictive comparisons. This gives the project a single analytical spine.

The first assumption is that the merged AIS and metocean table is the common source of truth. All downstream models use the same temporal and spatial alignment. This reduces the chance that two notebooks tell different stories because they were built from slightly different data versions. It also makes debugging easier: if a result changes, the team can check whether it came from feature engineering, modelling, or scenario logic.

The second assumption is that the rotor scenario should remain separate from the baseline speed model. This is a conservative design choice. It avoids training a model that silently mixes historical vessel behaviour with hypothetical rotor contribution. It also lets the same baseline prediction be compared before and after adding the rotor scenario. That makes the difference easier to explain in the report and presentation.

The third assumption is that the results should be read as decision-support signals, not as certified performance guarantees. The project uses real trajectory and environmental data, but it does not yet include the full onboard instrumentation needed for a formal vessel-performance trial. For a university project report, the useful contribution is the workflow: it shows how to connect AIS, weather, machine learning, rotor physics, and operational metrics in one traceable analysis.

These assumptions do not weaken the project if they are stated clearly. On the contrary, they make the results easier to trust because the reader can see the boundary between observed data, learned prediction, and simulated rotor effect. That boundary is important in maritime analytics, where operational decisions often depend as much on transparent reasoning as on raw model accuracy.

# References


[1] International Maritime Organization. *Fourth IMO Greenhouse Gas Study 2020*. IMO, 2020. `https://www.imo.org/en/ourwork/environment/pages/fourth-imo-greenhouse-gas-study-2020.aspx`

[2] International Telecommunication Union. *Recommendation ITU-R M.1371-6: Technical characteristics for VHF automatic identification system using time division multiple access in the maritime mobile service*. ITU, 2026. `https://www.itu.int/rec/R-REC-M.1371`

[3] H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, A. Horanyi, J. Munoz-Sabater, and others. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020. `https://doi.org/10.1002/qj.3803`

[4] Copernicus Marine Service. *Copernicus Marine Environment Monitoring Service: marine data and information*. European Union Copernicus Programme. `https://www.copernicus.eu/en/copernicus-services/marine`

[5] M. Traut, P. Gilbert, C. Walsh, A. Bows, A. Filippone, P. Stansby, and R. Wood. Propulsive power contribution of a kite and a Flettner rotor on selected shipping routes. *Applied Energy*, 113:362–372, 2014. `https://doi.org/10.1016/j.apenergy.2013.07.026`

[6] R. Lu and J. W. Ringsberg. Ship energy performance study of three wind-assisted ship propulsion technologies including a parametric study of the Flettner rotor technology. *Ships and Offshore Structures*, 15(3):249–258, 2020. `https://doi.org/10.1080/17445302.2019.1612544`

[7] T. Chen and C. Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794, 2016. `https://doi.org/10.1145/2939672.2939785`

[8] OpenStreetMap contributors. *OpenStreetMap*. `https://www.openstreetmap.org/copyright`